

CONSUMER SHOPPING BEHAVIOR ANALYSIS

Data Analytics Portfolio Project | Power BI

Item	Details
Dataset	Consumer Shopping Behavior & Buying Survey Dataset
Records	500 respondents
Tools	Power BI Power Query DAX
Analysis	Descriptive & Exploratory Analysis
Source	Kaggle - harpartapsingh13

Updated portfolio report reflecting the redesigned dashboard with donut, treemap, pie, bar, gauge, and clustered-column visuals.

Report Contents

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1. Project Overview

This project analyzes consumer shopping behavior using survey data from 500 respondents. The analysis focuses on shopping preferences, online-shopping habits, purchase decision factors, spending patterns, payment preferences, and customer experiences with online shopping.

The dataset was cleaned and transformed using Power Query, explored through multiple analytical views, and visualized using an interactive Power BI dashboard. The dashboard supports filtering by gender, age group, income range, and country.

The objective is to convert raw survey data into clear, decision-oriented insights that may support pricing, promotion, customer experience, payment, and omnichannel retail decisions.

2. Business Problem

Retail and e-commerce businesses need to understand not only where consumers shop, but also what influences their purchasing decisions, how they prefer to pay, how much they spend, and how they experience online shopping.

- Identify preferred shopping channels.
- Understand which factors receive the highest purchase-driver scores.
- Measure online-shopping frequency and identify frequent online shoppers.
- Examine non-essential spending bands and leading spending categories.
- Understand preferred payment methods.
- Assess online-shopping satisfaction, review trust, return behavior, and purchase regret.
- Explore selected relationships between behavioral variables without claiming causation.

3. Dataset Description

The dataset contains 500 survey responses across 30 original fields. It combines demographic attributes with shopping behavior, purchase drivers, spending and payment preferences, and online-shopping experience measures.

Data Area	Example Fields
Demographics	Age, Gender, Occupation Status, Country/Region, Monthly Income Range
Shopping Behavior	Shop Most Where, Online Shopping Frequency, Preferred Platform, Primary Device
Purchase Drivers	Review Influence, Social Ads Influence, Price Comparison, Discount Importance, Likelihood After Ad, Impulse Purchase
Spending & Payment	Top Spending Category, Monthly Non-Essential Spend, Preferred Payment Method, BNPL Usage
Customer Experience	Online Satisfaction, Trust in Reviews, Return Frequency, Purchase Regret

Most purchase-driver scores use a 1-5 scale, while online-shopping satisfaction and trust in online reviews use 1-10 scales. Monthly spending and income are provided as bands rather than exact monetary values.

Dataset source: <https://www.kaggle.com/datasets/harpartapsingh13/consumer-shopping-behavior-and-buying-survey-dataset>

4. Data Cleaning & Transformation

Data preparation was completed in Power Query before analysis. A raw query was retained and a separate cleaned query was used for transformation so that the original data remained available.

- Audited columns using quality, distribution, and profiling tools.
- Corrected data types, including timestamp and numeric score fields.
- Cleaned the Age field by retaining valid whole-number ages and converting invalid text entries to null.
- Created Age Group for demographic analysis.
- Applied Trim and Clean to text fields and standardized category labels.
- Represented missing categorical responses as Unknown where appropriate.
- Retained missing numerical survey scores as null rather than replacing them with zero.
- Created helper/sort columns for age groups, income bands, spending bands, and other ordinal categories.
- Checked complete duplicate rows and Response_ID uniqueness; no duplicate IDs were identified.

Why sorting columns were created

Power BI can sort text categories alphabetically even when the categories have a natural analytical order. Numeric helper columns were created so ordered categories such as Under \$50 -> \$50-150 -> \$150-300 -> \$300-500 -> \$500+ appear correctly in visuals.

5. Exploratory Data Analysis

EDA was organized into five dedicated Power BI pages:

1. Consumer Profile
2. Shopping Behaviour
3. Purchase Drivers
4. Spending & Payment
5. Customer Experience

Selected relationship analysis included Likelihood to Buy After Ad vs. Social Ad Influence, Price Comparison vs. Discount Importance, Online Satisfaction vs. Review Trust, and Purchase Regret vs. Return Frequency. These comparisons helped distinguish stronger patterns from relationships that appeared relatively weak or flat.

6. Power BI Dashboard

The redesigned dashboard combines headline KPIs with several visual forms and demographic slicers. This improves visual variety while retaining an analytical purpose for each chart.

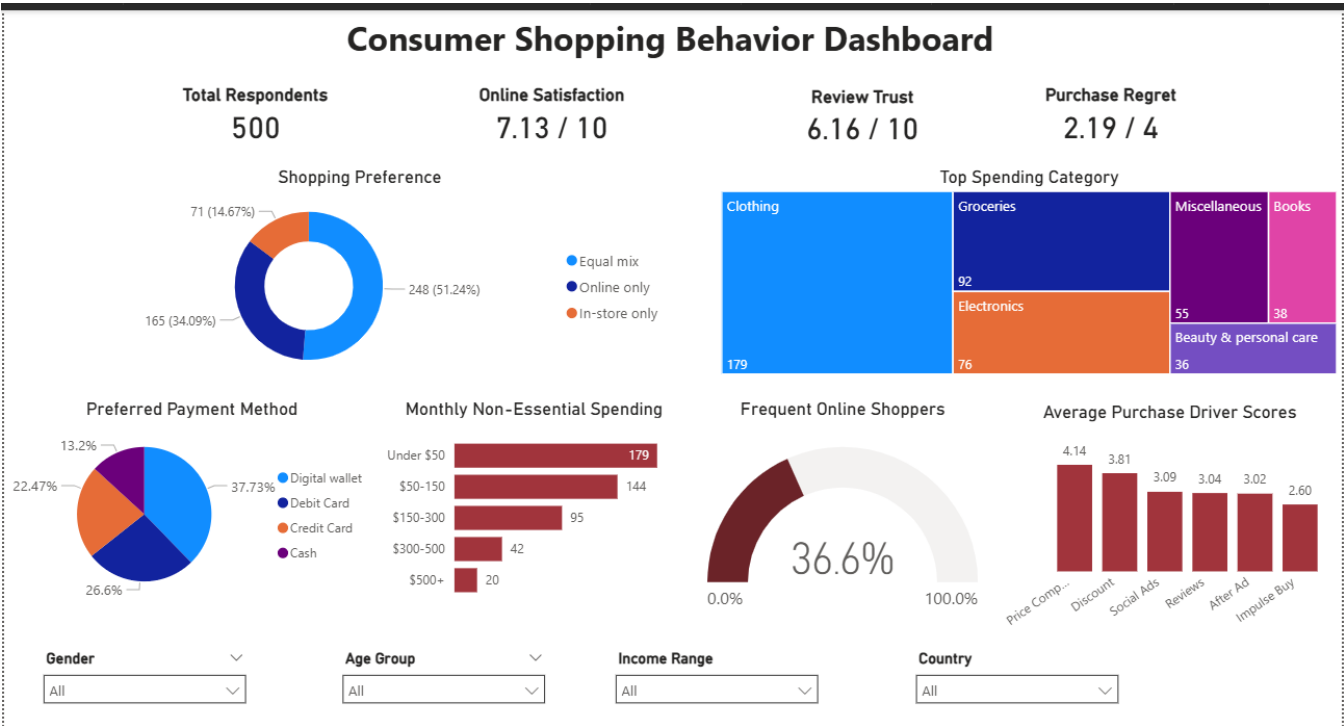


Figure 1. Final Consumer Shopping Behavior Dashboard

Dashboard KPIs

KPI	Value
Total Respondents	500
Online Satisfaction	7.13 / 10
Review Trust	6.16 / 10
Purchase Regret	2.19 / 4
Frequent Online Shoppers	36.6% of total sample

Dashboard Visuals

Analysis	Visual
Shopping Preference	Donut chart
Top Spending Category	Treemap
Preferred Payment Method	Pie chart
Monthly Non-Essential Spending	Horizontal bar chart
Frequent Online Shoppers	Gauge
Average Purchase Driver Scores	Clustered column chart

Interactive slicers: Gender, Age Group, Income Range, and Country.

7. Key Insights

1. Mixed-channel shopping is the largest valid preference group

Among the 484 respondents with a displayed shopping-preference response, 248 (51.24%) selected an equal mix of online and in-store shopping, 165 (34.09%) selected online only, and 71 (14.67%) selected in-store only.

2. Price-related factors have the highest purchase-driver averages

Price Comparison scored 4.14/5 and Discount Importance 3.81/5, above Social Ads Influence (3.09), Review Influence (3.04), Likelihood After Ad (3.02), and Impulse Purchase (2.60).

3. Frequent online shoppers represent 36.6% of the full sample

The gauge defines frequent online shoppers as respondents selecting Often or Always. There are 183 such respondents (132 Often + 51 Always), equal to 36.6% of all 500 respondents.

4. Digital wallets are the most frequently selected preferred payment method

Among displayed valid responses, Digital Wallet has 183 selections, followed by Debit Card (129), Credit Card (109), and Cash (64).

5. Non-essential spending is concentrated in lower bands

The largest groups are Under \$50 (179) and \$50-150 (144), followed by \$150-300 (95), \$300-500 (42), and \$500+ (20).

6. Clothing is the most frequently reported top spending category

Clothing leads with 179 respondents, followed by Groceries (92) and Electronics (76).

7. Online satisfaction exceeds review trust

Average online-shopping satisfaction is 7.13/10, compared with average review trust of 6.16/10.

8. Business Recommendations

- Support an omnichannel experience that combines online convenience with physical-store access.
- Emphasize competitive pricing and transparent value communication, while testing targeted discounts.
- Develop engagement and retention strategies for frequent and occasional online shoppers.
- Make digital-wallet checkout simple and reliable while retaining alternative payment options.
- Design offers for lower-to-mid non-essential spending bands while separately segmenting higher-spending customers.
- Give clothing appropriate merchandising attention for similar audiences while validating decisions with actual sales data.
- Strengthen review credibility through verified-purchase indicators and transparent review practices.

These recommendations are business hypotheses based on survey patterns and should be validated using transactional, profitability, campaign, and experimental data before implementation.

9. Tools & Technologies

Tool / Skill	Application
Power BI	Dashboard development and visualization
Power Query	Data cleaning and transformation
DAX	KPI and analytical measure creation
Microsoft Excel	Source dataset
EDA	Distribution and relationship analysis
Data Visualization	Business-focused visual design
Business Analysis	Translating analytical findings into decision-oriented insights

10. Limitations

- The survey sample contains 500 respondents and may not represent the wider consumer population.
- Responses are self-reported and may be affected by recall or response bias.
- Observed relationships are associations, not causal effects.
- Income and non-essential spending are provided as ranges rather than exact monetary values.
- Some survey fields contain missing responses; category-specific percentages may therefore use fewer than 500 valid responses.
- Recommendations should be validated using real transactional and business-performance data.

11. Conclusion

This project demonstrates an end-to-end Power BI data analytics workflow, from raw survey data through data cleaning, transformation, EDA, DAX measure development, and interactive dashboard design.

The analysis indicates that mixed online/in-store shopping is the largest valid shopping-preference group, price comparison and discounts receive the highest purchase-driver scores, digital wallets are the most frequently selected payment method among displayed valid responses, and lower non-essential spending bands contain the largest respondent groups. The redesigned dashboard makes these findings easier to explore through demographic slicers and multiple purpose-specific visual types.